Aryan Rastogi

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Education

Manipal University Jaipur — Jaipur, India
B.Tech in Computer Science and Engineering | CGPA: 8.94/10 — *Sept 2024 – May 2028*

Technical Skills

Languages: C++, Python, SQL, TypeScript
AI/ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, OpenCV, YOLOv8
Full Stack: React, Next.js, Node.js, FastAPI, REST APIs, Supabase
Tools & Platforms: AWS, Google Cloud Platform, Docker, Git, Neo4j
Techniques: Fine-tuning, RAG, Knowledge Graphs, NLP, Computer Vision, DSA

Work Experience

Hewlett Packard Enterprise — Jaipur, India
Career Preview Program – CPP-2 — *Sept 2026 – Present*
- Selected for HPE's pre-internship industry track; working on *AI-Based Autonomous System Validation & Reliability Checker*, a predictive disk-failure self-healing agent for server fleets under HPE mentor guidance.

Sylus AI — Remote
AI Engineer Intern — *April 2026 – Sept 2026*
- Engineered the platform end-to-end from a greenfield repository to production, owning auth, database schema, and core architecture through 180+ commits, then scaling it into a multi-tenant AI content platform.
- Designed and shipped a 9-phase agency system (RBAC, approval workflows, client portal, invites, real-time collaboration) from scratch, taking the product from single-user to multi-brand agency-ready.
- Built the AI content engine (multi-model LLM copywriting, live Google News trend discovery, AI image generation) and a document intelligence pipeline auto-extracting and scheduling posts from PDFs, Excel, and Word docs.
- Hardened the platform for production: patched security vulnerabilities across auth, API endpoints, and middleware (SSRF, CORS), and stabilized deployment across Cloud Run and Docker.

Southern Federal University — Russia (Remote)
SDE Intern — *Oct 2025 – Nov 2025*
- Developed a web application as part of an international interdisciplinary project with SFedU and MUJ to help international students navigate and adapt to campus facilities.

Elumina Street Solutions — Remote
AI Research & Development Intern — *March 2025 – June 2025*
- Developed a real-time violence detection model combining CNN-MobileNet and LSTM in TensorFlow; improved class imbalance handling with focal loss and class weighting.
- Implemented and evaluated a real-time weapon detection system using YOLOv8 and OpenCV, collaborating with the research team on hyperparameter optimization.

AI2: Computational Linguistics & Vision Lab — Jaipur, India
Research Intern — *Jan 2025 – Present*
- Investigated safety-creativity trade-offs in LLMs; findings contributed to *Quantifying the Alignment Tax: Tracking Creativity and Semantic Entropy through the SFT and DPO Pipeline*.
- Designed taxonomy-guided safety evaluation pipeline across multiple alignment paradigms (SFT, LoRA, DPO, RLHF); contributed to *SafeAlign: A Multi-Paradigm Pipeline for LLM Safety Evaluation via Taxonomy-Guided Data Curation*.

Projects

TeXlab – Prompt-to-LaTeX Diagram Pipeline | *texlab.aryy.in*
- Architected a staged generation pipeline (type router, structured JSON spec, algorithmic layout, component assembly) converting natural language prompts into publication-ready LaTeX diagrams with Overleaf integration and self-repairing compilation loops; router reached 97% accuracy on held-out prompts.

RoadEye – AI-Based Road Safety Audit System | *IIT Madras NRSH 2025*
- Built an end-to-end AI pipeline for automated road infrastructure auditing using YOLO-based object detection to identify road assets and defects, generating structured PDF audit reports with risk scores and GIS-ready outputs.

UniWay – Campus Navigation Platform | *uniway.aryy.in* | *Hackathon Winner*
- Developed a campus navigation platform with interactive maps, smart search, and route guidance using React, TypeScript, Leaflet, and Google Maps API; won 1st place and led to international research collaboration with SFedU, Russia.

Achievements

National Road Safety Hackathon (IIT Madras): Ranked in top 10 finalist teams nationwide out of 3,500+ competing teams; presented RoadEye to a panel from CoERS, IIT Madras.
UniWay Hackathon – 1st Place: Won first place and secured an interdisciplinary research collaboration with Southern Federal University, Russia.
Student Excellence Award: Received consecutively in the 3rd and 4th semesters at MUJ for outstanding academic and extracurricular performance.